Homework #3

This is the homework for week 3. The due date is 11:59 p.m., September 29. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1**): (50 points) Suppose $X_1, X_2, ..., X_n \overset{iid}{\sim} Pois(\lambda_0)$, i.e., Poisson distribution with parameter λ,

$$Prob(X_1 = k) = \frac{\lambda^k}{k!}\exp(-\lambda), k \geq 0.$$

(a) Calculate the expectation $\mathbf{E}X_1$. Based on this, find the method of moments estimator for λ_0.

(b) Derive the bias, the variance, and the MSE of the method of moment estimator.

(c) Estimate the probability $Prob(X_1 > 5)$. Notably, λ_0 is unknown.

(d) Find the maximum likelihood estimator for λ_0.

(e) Find the maximum likelihood estimator for $\log(\lambda_0)$.

(f) Show your maximum likelihood estimator for $\log(\lambda_0)$ is consistent.

(**Question2**): (30 points) This problem shows what happen when the probability model is not correctly specified. Suppose $X_1, X_2, ..., X_n \overset{iid}{\sim} Ga(a_0, b_0)$, i.e., the density

$$f(x) = \frac{b_0^{a_0}}{\Gamma(a_0)}x^{a_0-1}\exp(-b_0 x),$$

but the modeller works with the (incorrect) Exponential model:

$$X_1, X_2, ..., X_n \overset{iid}{\sim} Exp(\lambda)$$

with some unknown true λ_0 in mind.

(a) Based on the exponential assumption, derive the method of moment estimator $\widehat{\lambda}$ for the hypothetical λ. Derive the expectation of $1/\widehat{\lambda}$ with the correct distribution. That is, $\mathbf{E}(1/\widehat{\lambda})$ should be a function of a_0 and b_0.

(b) Find the maximum likelihood estimator for λ, denoted as $\hat{\lambda}$.

(c) Discuss if and when the estimator $\hat{\lambda}$ is consistent.

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

i. Generate 500 i.i.d. random variables $X_1, \cdots, X_{500}$ with uniform distribution on $[0, 1]$.

ii. Calculate and record the running sample means:

$$\frac{1}{t}\sum_{i=1}^{t} X_i$$

for $t = 1, 2, ..., 500$.

iii. Repeat the experiment for 10 times, plot the running sample means in the same figure. What can you observe from the numerical exercises?